

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VIII

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Class :SYIT	Batch : 01
Date of Assignment :12/08/26	Date/Time of Submission :12/08/26

Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

- a. Bubble sort

Code:

```
practical7.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/practical7.py (3.14.6)
File Edit Format Run Options Window Help
def bubblesort(array):
    for i in range(len(array)):
        for j in range(0, len(array)-i-1):
            if array[j]>array[j+1]:
                temp =array[j]
                array[j]=array[j+1]
                array[j+1]=temp

data=[-2,4,0,11,-9,12,99,-25]
bubblesort(data)

print('sorted Array in Ascending Order:')
print(data)
|
```

Output:

```
=====
sorted Array in Ascending Order:
[-25, -9, -2, 0, 4, 11, 12, 99]
>>>
```

b. Insertion sort

Code:

```
practical7.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/practical7.py (3.14.6)
File Edit Format Run Options Window Help
def insertionSort(array):
    for step in range(1,len(array)):
        key = array[step]
        j = step - 1

        while j >=0 and key < array[j]:
            array[j+1]=array[j]
            j= j-1

        array[j+1] = key

data = [9,12,8,6,4,3,0]
insertionSort(data)
print("Sorted Array in Ascending Order:")
print(data)
```

Output:

```
=====
Sorted Array in Ascending Order:
[0, 3, 4, 6, 8, 9, 12]
>>>
```

c. Selection sort

Code:

```
practical7.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/practical7.py (3.14.6)
File Edit Format Run Options Window Help
def selectionSort(array, size):

    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            if array[i] < array[min_idx]:
                min_idx = i

        (array[step], array[min_idx]) = (array[min_idx], array[step])

data = [-2, 45, 0, 11, 6, 12, 22, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Ascending Order:')
print(data)
```

Output:

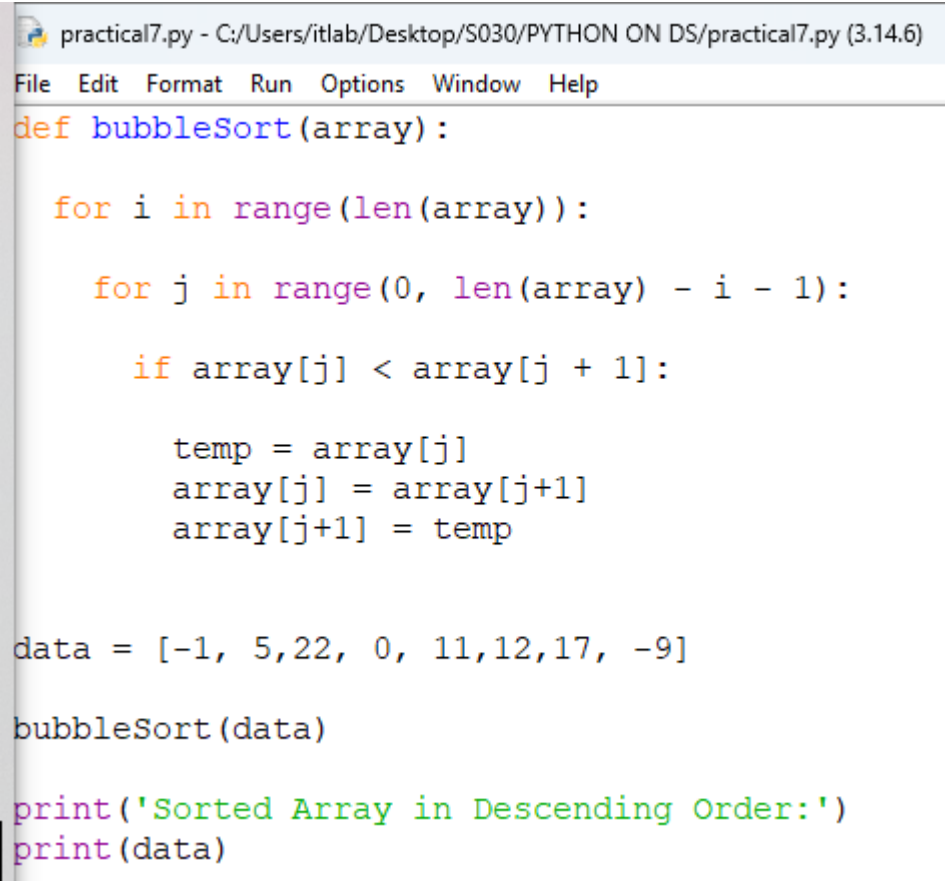
```
==== RESTART: C:/Users/itlab/Desktop/
Sorted Array in Ascending Order:
[-9, -2, 0, 6, 11, 12, 22, 45]
>>>
```

Curiosity Questions:

Programming Questions

1. Write a Python program to sort a list of numbers in descending order using **Bubble Sort**.

Code:

A screenshot of a Python IDE window titled 'practical7.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/practical7.py (3.14.6)'. The window has a menu bar with 'File', 'Edit', 'Format', 'Run', 'Options', 'Window', and 'Help'. The code is written in a syntax-highlighted font. It defines a 'bubbleSort' function that takes an 'array' as input. The function uses two nested loops: an outer loop 'for i in range(len(array)):' and an inner loop 'for j in range(0, len(array) - i - 1):'. Inside the inner loop, there is an 'if array[j] < array[j + 1]:' condition. If true, it swaps the elements at indices 'j' and 'j+1' using a temporary variable 'temp'. After the function definition, a list 'data = [-1, 5, 22, 0, 11, 12, 17, -9]' is created. The function is called with 'bubbleSort(data)'. Finally, two print statements are shown: 'print('Sorted Array in Descending Order:')' and 'print(data)'.

```
practical7.py - C:/Users/itlab/Desktop/S030/PYTHON ON DS/practical7.py (3.14.6)
File Edit Format Run Options Window Help
def bubbleSort(array):
    for i in range(len(array)):
        for j in range(0, len(array) - i - 1):
            if array[j] < array[j + 1]:
                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp

data = [-1, 5, 22, 0, 11, 12, 17, -9]

bubbleSort(data)

print('Sorted Array in Descending Order:')
print(data)
```

Output:

2. Write a Python program to sort a list of numbers in descending order using **Insertion Sort**.
3. Write a Python program to sort a list of numbers in descending order using **Selection Sort**.